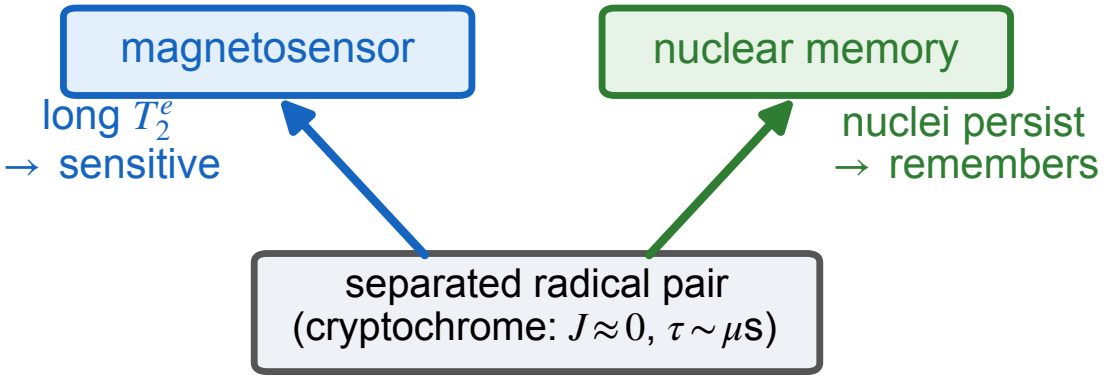
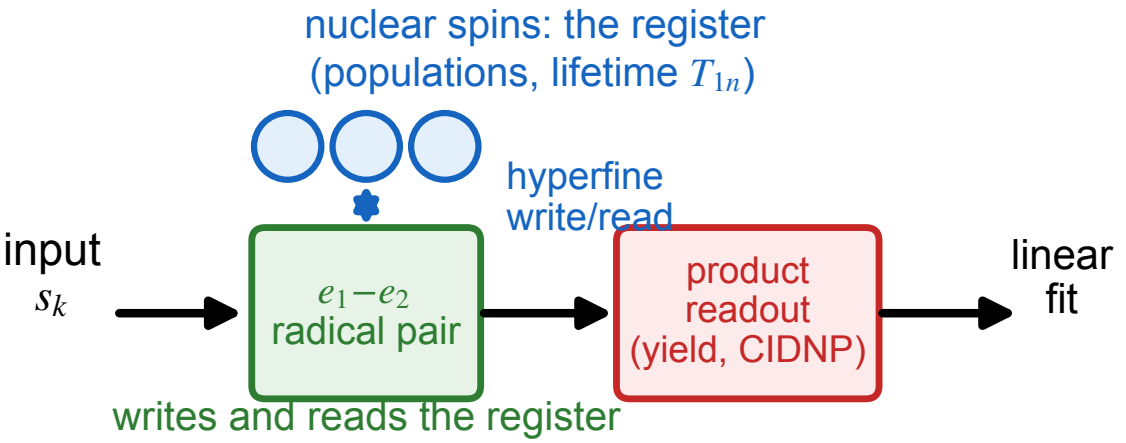
(a) one molecule, two timescales



each cycle discards the electron pair and keeps the nuclei

(b) nuclei are the register that must persist



$$\text{memory horizon} = (\mathrm{MC} - C_0) \times T_{\text{turnover}}$$

outlasts a turnover only if the register is reused *and* survives:

proton-carried (^{14}N is quadrupolar); reaching 10 ms needs $\tau_{\text{eff}} \lesssim 1.0$–$6.5$ ns and a catalytic, not light-driven, turnover